

CT Dose Reduction and Dose Management Tools: Overview of Available Options¹

TEACHING POINTS

See last page

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In the past decade, the tremendous advances in computed tomography (CT) technology and applications have increased the clinical utilization of CT, creating concerns about individual and population doses of ionizing radiation. Scanner manufacturers have subsequently implemented several options to appropriately manage or reduce the radiation dose from CT. Modulation of the x-ray tube current during scanning is one effective method of managing the dose. However, the distinctions between the various tube current modulation products are not clear from the product names or descriptions. Depending on the scanner model, the tube current may be modulated according to patient attenuation or a sinusoidal-type function. The modulation may be fully preprogrammed, implemented in near-real time by using a feedback mechanism, or achieved with both preprogramming and a feedback loop. The dose modulation may occur angularly around the patient, along the long axis of the patient, or both. Finally, the system may allow use of one of several algorithms to automatically adjust the current to achieve the desired image quality. Modulation both angularly around the patient and along the z-axis is optimal, but the tube current must be appropriately adapted to patient size for diagnostic image quality to be achieved.

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Abbreviations: AEC = automatic exposure control, ALARA = as low as reasonably achievable

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Teaching Point

Table 1
Mechanisms for Dose Reduction at CT

X-ray beam filtration
X-ray beam collimation
X-ray tube current modulation and adaptation for patient body habitus (automatic exposure control)
Peak kilovoltage optimization
Improved detection system efficiency
Noise reduction algorithms

Introduction

Unlike screen-film radiographs, images acquired with computed tomography (CT) never look overexposed in the sense of being too dark or too light. The normalization of CT data to represent a fixed amount of attenuation relative to that of water ensures that the image always appears properly exposed. Thus, CT users have not been compelled to decrease the tube current–time product or the peak kilovoltage for scanning in small patients, and, as a result, such patients often were exposed to an excessive radiation dose. In addition, image quality changes noticeably (specifically, with regard to the amount of noise or graininess) according to the patient's body habitus. For scanning in large patients, the dose must be increased to obtain diagnostic-quality images. Thus, there has been a tendency to increase the tube current–time product (patient dose) to avoid excessive noise on images, particularly for large patients and at thin-section CT, which is more readily available with the newer generations of scanners. **As the growth in CT utilization increased, particularly in pediatric patients, and as concern about the population dose from CT began to be expressed in the scientific literature and lay press (1–4), it became clear that the responsible use of CT required an adjustment of technique factors on the basis of patient size (attenuation characteristics) (2,5,6).** In response to these concerns, the radiology community (radiologists, medical physicists, and manufacturers) implemented CT dose management procedures that correspond to the principle of ALARA (as low as reasonably achievable) (3,7–13). The guiding principle in selecting the right dose for a CT examination is that the specific patient attenuation and the specific diagnostic task must be taken into account. For large patients, a dose that is higher than that for small patients is consistent with the ALARA principle. Our objective in this article is to educate users of CT scanners about the dose management and dose reduction tools available with current CT systems.

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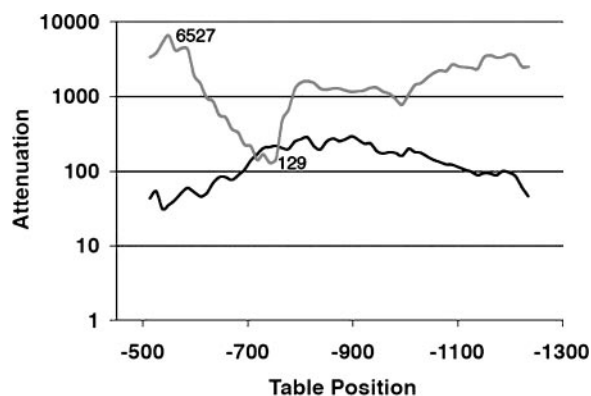


Figure 1. Graph of relative attenuation values (top) according to table position (in millimeters from the initial position) and associated body region (bottom) shows strong variations in attenuation, by almost three orders of magnitude, according to body region and projection angle. Relative attenuation values, which reflect the sum of x-ray absorption along an imaginary line through the patient, vary with gantry rotation from the anteroposterior direction (black line) to the lateral direction (gray line).

In 1981, Haaga et al (14) published the concept of using tube current variation to reduce the radiation dose while maintaining image quality. In 1994, GE Medical Systems made available the first commercial tube current modulation system, with which the dose could be reduced by as much as 20% (15). In 1993 and 1994, Kalender et al (16,17) reported dose reductions of up to 40% in elliptical body regions, reductions achieved by using anatomically based modulation of the tube current. Additional tube current modulation products became available in late 2001, when, partly because of the public concerns about dose, dose reduction became a marketing tool. This development left the radiology community with the need to sort through the various trade names for a wide variety of dose reduction products. **While there are additional technical mechanisms for dose reduction at CT (Table 1), this review is focused only on tube current modulation and patient size–dependent tube current adaptation, two mechanisms that are jointly referred to as automatic exposure control (AEC).** A summary of the generic types of tube current modulation, as well as specific details about currently available products, is given to help CT users understand the differences among CT dose reduction products. In addition, we discuss the various paradigms used in AEC systems by the manufacturers to allow users to select the desired image quality,

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Table 2
Angular Tube Current Modulation Systems

Characteristic	GE Healthcare	Philips	Siemens	Toshiba
Product name	Smart Scan*	DOM (Dose Modulation)	Care Dose	...
Requires CT projection radiograph	No	No	No	...
Modulation algorithm	Sinusoidal	Attenuation based	Attenuation based	...
Online feedback	No	Yes	Yes	...

*This system is available only on the CT/i scanner model.

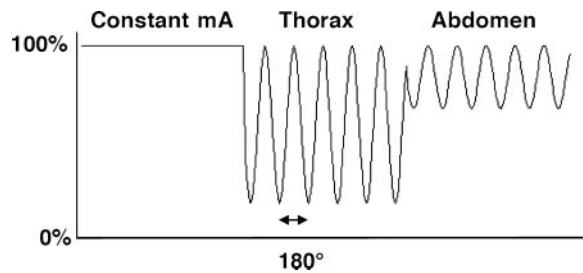


Figure 2. Illustration of the concept of angular dose modulation, in which the tube current (mA) (vertical axis) is varied as the x-ray tube rotates around the patient (horizontal axis), shows greater variation in the thorax than in the abdomen.

and we discuss the determination of optimal image quality. Finally, we review our clinical experience with a commercial AEC system.

X-ray Tube Current Modulation

Extremely large variations in patient radiation absorption occur with variations in projection angle and anatomic region (Fig 1). Since the projection with the most noise primarily determines the amount of noise on the final image, it is possible to reduce the dose (photons) for other projections without increasing the noise on the final image (14,17). In addition, it is also possible to reduce the dose for projections that are of limited interest. For example, by decreasing the tube current during systole, the scanner can reduce the dose for a cardiac CT examination by as much as 50% (18).

Angular Tube Current Modulation

Angular (x- and y-axis) tube current modulation involves variation of the tube current to equalize the photon flux to the detector as the x-ray tube rotates about the patient (eg, from the anteroposterior direction to the lateral direction). The operator chooses the initial value for the tube current–time product, and the tube current is modulated (typically, decreased) from the initial value within one gantry rotation (Fig 2).

Kalender et al (17) demonstrated a decrease in streak artifacts on images at the level of the shoulders when the tube current was increased so that the projection noise level was more uniform among anteroposterior and lateral projections. In some implementations of angular dose modulation, an increase in tube current may be allowed in the shoulder region, whereas in other implementations the tube current is not allowed to exceed the initial value prescribed by the operator. As the x-ray tube rotates among the anteroposterior and lateral positions, the tube current can be varied sinusoidally according to the attenuation information on the CT projection radiograph, or in near-real time according to the measured attenuation from the previous 180° projection. Details regarding the implementation of angular dose modulation by several manufacturers are given in Table 2.

Longitudinal Tube Current Modulation

Longitudinal (z-axis) tube current modulation involves variation of the radiation dose among anatomic regions (eg, shoulders vs abdomen vs pelvis) by varying the tube current along the z-axis of the patient. Unlike angular tube current modulation, in which the tube current is varied cyclically in relation to the starting tube current value, the task of z-axis modulation is to produce relatively uniform noise levels across the various regions of the anatomy. Thus, the operator must select the desired level of image quality for input to the algorithm by using one of the following manufacturer-specific methods: the reference noise index (GE Healthcare Technologies, Waukesha, Wis), reference image acquisition (Philips Medical Systems, Best, the Netherlands), reference tube current–time product value (Siemens Medical Solutions, Forchheim, Germany), or reference standard deviation or image quality level (Toshiba Medical Systems, Tokyo, Japan).

The tube current is modulated to provide the desired level of image quality as the attenuation

Table 3
Longitudinal Tube Current Modulation Systems

Characteristic	GE Healthcare	Philips	Siemens	Toshiba
Product name	Auto mA	...	...	Real E.C.
Requires CT projection radiograph	Yes	...	...	Yes
Modulation algorithm	Attenuation based	...	...	Attenuation based
Online feedback	No	...	...	No

varies between anatomic regions (Fig 3). Details regarding implementation by several manufacturers are given in Table 3.

Angular-Longitudinal Tube Current Modulation

The simultaneous combination of angular and longitudinal (x-, y-, and z-axis) tube current modulation involves variation of the tube current both during gantry rotation and along the z-axis of the patient (ie, from the anteroposterior direction to the lateral direction, and from the shoulders to the abdomen). The operator must still indicate the desired level of image quality by one of the methods described earlier. This is the most comprehensive approach to CT dose reduction because the x-ray dose is adjusted according to the patient-specific attenuation in all three planes. Details regarding the implementation of this dose modulation technique by several manufacturers are given in Table 4. A graphic illustration of this approach is shown in Figure 4.

Automatic Exposure Control

AEC is analogous to acquisition timing in general radiography. The user determines the image quality requirements (as regards noise or the contrast-to-noise ratio), and the CT system determines the right tube current–time product. In practice, it is relatively straightforward for the system to deliver the desired image quality, once that has been defined. However, it can be quite difficult to achieve agreement on the image quality requirement for the various CT examination types and patient age groups.

In defining the required image quality, the user needs to remember that pretty pictures are not needed for all diagnostic tasks, but, rather, a choice can be made between low noise and a low dose, depending on the diagnostic task. The CT system will then adjust the tube current during the gantry rotation, during movement along the

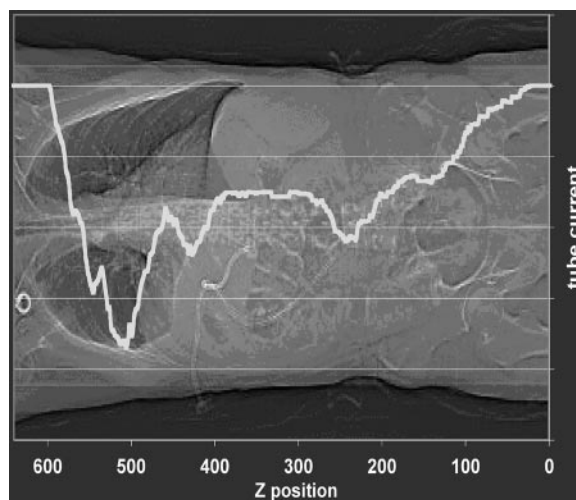


Figure 3. Graph of tube current (in milliamperes) superimposed on a CT projection radiograph illustrates the concept of longitudinal dose modulation, with variation of the tube current along the z-axis. The curve is determined by using attenuation data from the CT projection radiograph and the manufacturer-specific algorithm.

z-axis, or during movement in all three dimensions, according to the patient's body habitus and the user's image quality requirements. Thus, we differentiate between the modulation of the tube current to achieve a defined image quality, and the prescription of the desired image quality by the user. Together these tasks are referred to as AEC.

Image Quality Selection Paradigms

Each manufacturer of CT systems uses a different method of defining the image quality in the user interface. However, the reference value, index, or image can be stored with a specific protocol in all manufacturer-implemented AEC systems.

GE Healthcare uses a concept known as the noise index. The noise index is referenced to the standard deviation of CT numbers within a region of interest in a water phantom of a specific

Table 4
Combined Angular and Longitudinal Tube Current Modulation Systems

Characteristic	GE Healthcare	Philips	Siemens	Toshiba
Product name	Smart mA	Z-DOM	CareDose 4D	Sure Exposure
Requires CT projection radiograph	Yes	Yes	Yes	Yes
Modulation algorithm	Attenuation based	Attenuation based	Attenuation based	Attenuation based
Online feedback	No	Yes	Yes	No

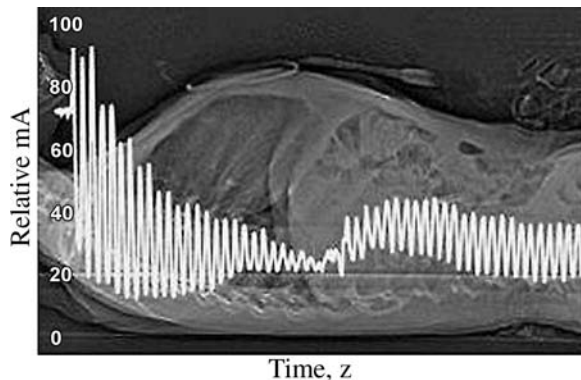


Figure 4. Graph of tube current (*mA*) superimposed on a CT projection radiograph shows the variation in tube current as a function of time (and, hence, table position along the *z*-axis) at spiral CT in a 6-year-old child. An adult scanning protocol and an AEC system (CareDose 4D; Siemens Medical Solutions) were used with a reference effective tube current–time product of 165 mAs. The mean effective tube current–time product for actual scanning was 38 mAs (effective tube current–time product = tube current–time product/pitch).

size. A lookup table is used to map the patient-specific attenuation values measured on the CT projection radiograph (“scout” image) to tube current values for each gantry rotation according to a proprietary algorithm. The algorithm is designed to maintain the same image noise level as the attenuation values change from one rotation to the next. Different noise indexes may be required for patients of different size (19).

Philips uses a reference image to help users select the desired image quality to be matched. In a process that the manufacturer calls automatic current setting, the user selects an acceptable patient examination, and the system saves the image data (including the raw CT projection data and the CT projection radiograph, or “surview”) as the reference data for comparison with the CT projection radiograph and data obtained from other patients in examinations for the same diag-

nostic task. The comparison is performed with use of a proprietary algorithm and on a protocol-by-protocol basis (ie, for a given examination type) to enable the automated selection of the appropriate tube current values by the scanner.

Siemens uses a “quality reference mAs” to help users establish the desired image quality level. For each examination type (ie, protocol), the user selects the effective tube current–time product (tube current–time product/pitch) typically used for CT in a patient with a weight of approximately 80 kg. (For pediatric protocols, the effective tube current–time product that should be selected is that typically used for CT in a 20-kg patient.) The noise target (standard deviation of CT numbers) is varied on the basis of patient size by using an empirical algorithm; thus, image noise is not kept constant for all patients but is adjusted according to an empirical impression of image quality. CT projection radiographs (“topograms”) for each patient are used to predict the tube current curve (with variations along the *x*-, *y*-, and *z*-axes) that will yield the desired image quality, given the patient’s size and anatomy. An online feedback system fine-tunes the actual tube current values during scanning to precisely match the patient-specific attenuation values at all angles (as opposed to the attenuation values estimated on the basis of the one angle of the CT projection radiograph).

Toshiba offers the user two methods of selecting the desired image quality: use of the standard deviation of CT numbers, or use of an image quality level. Both methods are referenced to the standard deviation of CT numbers measured in a patient-equivalent water phantom. Data from the patient’s CT projection radiograph (“scanogram”) are used to map the selected image quality to tube current values.

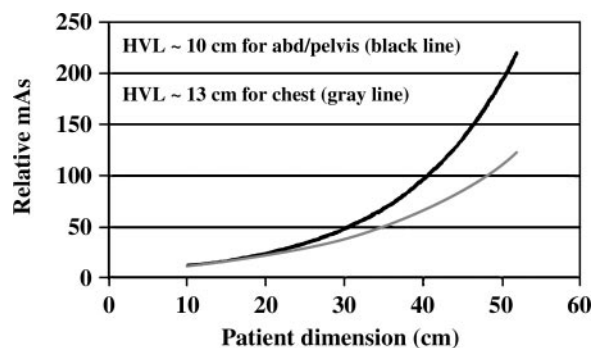


Figure 5. Curves show the relationship between the tube current–time products (in milliamperere-seconds) that are used for abdominal (*abd*) and pelvic versus chest CT examinations at the authors' institution and the patient's lateral width (in centimeters) at the level of the liver (20). *HVL* = half-value layer.

Determination of the Optimal Image Quality

Image quality is a nonspecific and subjective measure of the readability of an image, which must be assessed by a trained observer. Objective measures such as image noise or contrast-to-noise ratio can be obtained relatively easily but may not completely capture all the features relevant to making a correct clinical diagnosis. Thus, the determination of optimal image quality can be a complex task, as both quantitative metrics (eg, noise) and observer perceptions are involved. A simplified approach to achieving the optimal image quality is to require a specific noise level for a specific diagnostic task.

Table 5 indicates noise measurements for scanning with a constant tube current–time product (chosen to be 130 mAs) in water phantoms of different diameter. Table 6 demonstrates the tube current–time product required to yield constant image noise (chosen to be 13.0 HU) as water phantom diameter again was varied. Tables 5 and 6 together demonstrate that it is not technically feasible to maintain constant image noise over all patient sizes, even if this were clinically desirable, because CT systems cannot achieve such extremely low and high tube current–time product values. The large range of tube current–time product values required to maintain constant image noise as the size of the subject varies is a con-

Table 5
Measured Noise at CT in Water Phantoms with Varied Diameter and a Constant Tube Current–Time Product of 130 mAs

	Diameter of Phantom (cm)					
Measurement	10	14	20	25	30	40
Noise (HU)	1.9	3.5	5.1	8.2	13.0	33.6

Table 6
Tube Current–Time Product Required to Maintain Constant Image Noise of 13 HU at CT in Water Phantoms with Varied Diameter

	Diameter of Phantom (cm)					
Measurement	10	14	20	25	30	40
Tube Current–Time Product (mAs)	2.9	6.2	19	50	130	869

sequence of the exponential nature of x-ray absorption.

While the avoidance of both extremely low and extremely high tube current–time product values is necessary with regard to the x-ray generator, we also found it most appropriate from a patient dose and image quality perspective. More aggressive dose reduction was not acceptable in children, and more aggressive dose increase was not necessary in obese patients. A similar experience was reported by Wilting et al (5) for images with constant noise that were obtained in patients of different size (ranging from small children to obese adults). The quality of the images obtained in pediatric patients was found unacceptable, even though the level of image noise was the same as that for the images of adult patients. Kalra et al (19) observed a similar situation when using the GE Healthcare noise index paradigm, which, for a given noise index, is intended to deliver constant noise across anatomic regions and patient sizes. They found that readers required a lower noise index (less image noise) for smaller patients and accepted a higher noise index (more image noise) for larger patients.

Although images with less noise may be required for small patients, a dramatic reduction of tube current–time product is still appropriate,

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Table 7
Effective Tube Current–Time Product
Decreases Relative to Values in Technique
Charts

Level	Decrease (%)*
All levels	21.0
Upper lung	29.7
Breast	54.8
Liver	13.2
Pelvis	23.2

*Indicates decrease in effective tube current–time product with use of the described AEC system, compared with the values that would have been used according to our technique charts, averaged over all anatomic levels and for four individual levels.

Table 8
Effective Tube Current–Time Product
Decreases Relative to a Constant Value Used
for All Adult Patients

Category	Decrease (%)*
All patients	18.5
Slim patients	44.9
Large patients	3.1

*Indicates decrease in effective tube current–time product with use of the described AEC system, compared with a single effective tube current–time product used for all adult patients.

given the decreased attenuation values. Our CT technique charts, which were developed with considerable clinical input from adult and pediatric radiologists, prescribe a four- to fivefold decrease in tube current–time product for infants (with resultant image noise that is still less than that for a standard adult) and a twofold increase in tube current–time product for obese patients (with image noise that is greater than that for a standard adult) (19). The scanner operator must measure the patient's lateral width on the CT projection radiograph and enter the appropriate tube current–time product value for each patient. A plot of the relationship between tube current–time product and patient width in our clinical practice (20) is shown in Figure 5. However, if consistent

compliance is unachievable in daily practice, the benefits of technique charts are lost.

Thus, we believe that a constant noise level for patients of all sizes is not the clinical ideal. Although radiologists are accustomed to “reading through the noise” on images obtained in large patients, most strongly prefer less image noise on images obtained in children, perhaps because most children do not have the fat planes between tissues and organs that are typical in adults and that enhance contrast and tissue differentiation. In addition, because the details of interest are smaller in children, higher contrast-to-noise ratios are required. A higher level of diagnostic confidence also appears to be demanded for pediatric imaging.

Clinical Experience with a Commercial AEC System

In our clinical practice, we have implemented an AEC system that uses modulation along the x-, y-, and z-axes (CareDose 4D; Siemens Medical Solutions, Forchheim, Germany). In the initial 4 months of use (from November 2003 through February 2004), 2200 patients were scanned with this system. The average adult patient in our practice was considered to have a lateral width of 35–40 cm. Patients with a lateral width of less than 35 cm were considered slim, and those with a lateral width of more than 40 cm were considered large. All width measurements were obtained at the level of the liver by using digital calipers and the CT projection radiograph. A log was kept of the effective tube current–time product that would have been used according to our technique chart for the specific protocol (ie, chest vs pelvic CT), the actual effective tube current–time product values used by the AEC system at four anatomic levels, and the mean effective tube current–time product value used by the AEC system over the entire examination. During this time period, more than 30 radiologists were assigned to use this scanner. Image quality was deemed equivalent to that achieved with use of the tube current–time products prescribed by the technique charts. Our dose reduction results are summarized in Tables 7 and 8. Clinical examples are shown in Figures 6 and 7.

Figure 6. Clinical CT examination of the abdomen and pelvis. **(a)** CT projection radiograph (topogram). **(b, c)** Axial CT images at the levels of the liver **(b)** and pelvis **(c)**. The patient's lateral width was 30 cm (measured from the topogram at the level of the liver). The reference effective tube current–time product for 5-mm-thick sections was 240 mAs, and the effective tube current–time product that would have been used according to the institutional technique chart was 120 mAs. The AEC system automatically adapted to the small patient size, as shown by the image quality in **b** (actual effective tube current–time product, 88 mAs) and **c** (actual effective tube current–time product, 122 mAs).



a.



b.



c.

Conclusions

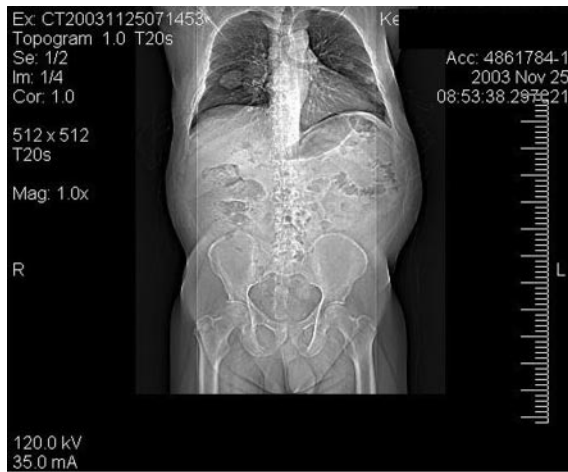
AEC systems in which the tube current is modulated along the x-, y-, and z-axes and in which the acceptable level of image noise is varied according to patient size, anatomic region, and diagnostic task can provide significant levels of dose reduction with minimal operator intervention. Even systems that offer only a subset of these features provide meaningful levels of dose reduction. **In the future, we expect that the use of AEC systems will become mandatory and that such systems will be available on all CT scanners. We strongly encourage users to take advantage of these technical mechanisms for reducing radiation dose while maintaining diagnostic image quality.**

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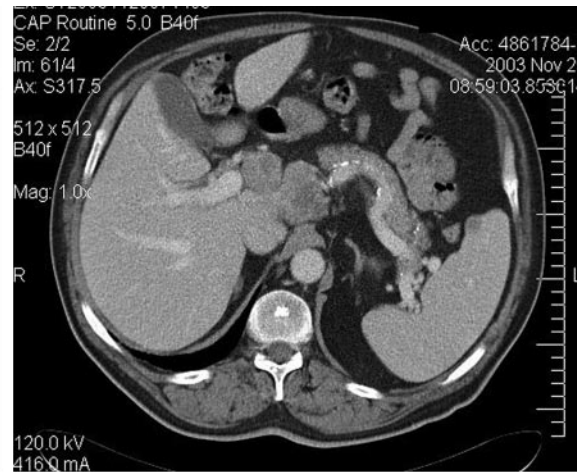
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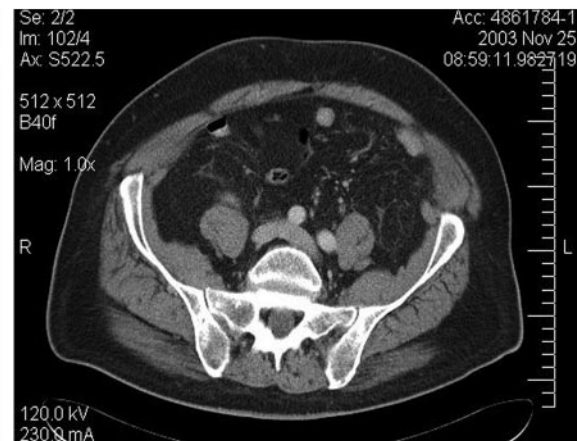
a.



d.



b.



e.



c.

Figure 7. Clinical CT examination of the chest, abdomen, and pelvis. **(a)** CT projection radiograph (topogram). **(b–e)** Axial CT images at the levels of the upper thorax **(b)**, middle thorax **(c)**, liver **(d)**, and pelvis **(e)**. The patient's lateral width was 43 cm (measured from the topogram at the level of the liver). Reference effective tube current–time product for 5-mm-thick sections was 240 mAs, and the effective tube current–time product that would have been used according to the institutional technique chart was 340 mAs. Actual effective values used by the AEC system were 95 mAs **(b)**, 101 mAs **(c)**, 369 mAs **(d)**, and 205 mAs **(e)**. The AEC system automatically adapted both to the larger patient size and the lower-attenuating regions of the body, as shown by the comparable image quality in **b–e**.

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Teaching Points for CT Dose Reduction and Dose Management Tools: Overview of Available Options

Cynthia H. McCollough, PhD, et al

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As the growth in CT utilization increased, particularly in pediatric patients, and as concern about the population dose from CT began to be expressed in the scientific literature and lay press (1–4), it became clear that the responsible use of CT required an adjustment of technique factors on the basis of patient size (attenuation characteristics) (2,5,6).

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While there are additional technical mechanisms for dose reduction at CT (Table 1), this review is focused only on tube current modulation and patient size–dependent tube current adaptation, two mechanisms that are jointly referred to as automatic exposure control (AEC).

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X-ray beam filtration
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X-ray tube current modulation and adaptation for patient body habitus (automatic exposure control)
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In the future, we expect that the use of AEC systems will become mandatory and that such systems will be available on all CT scanners. We strongly encourage users to take advantage of these technical mechanisms for reducing radiation dose while maintaining diagnostic image quality.